

Experiment No. 1

ARM IDE's, Toolchain and LPC2148 board

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Introduction to ARM IDEs, Development Toolchain and LPC2148 Evaluation Board

1.1 Course Outcome Addressed

CO1: Identify an appropriate hardware and software architecture for effective design and development of an embedded system with optimized design metrics.

1.2 Aim

To study the ARM embedded software development environment, ARM development toolchain and LPC2148 evaluation board, and configure an ARM project for embedded system development.

1.3 Objectives

After completing this experiment, students will be able to:

- Understand the ARM embedded software development environment.
- Study the ARM software development toolchain.
- Identify the hardware features of LPC2148 evaluation board.
- Understand onboard peripherals available on LPC2148.
- Configure an ARM project using Keil μ Vision IDE.
- Select appropriate hardware and software architecture for embedded system design.

1.4 Theory

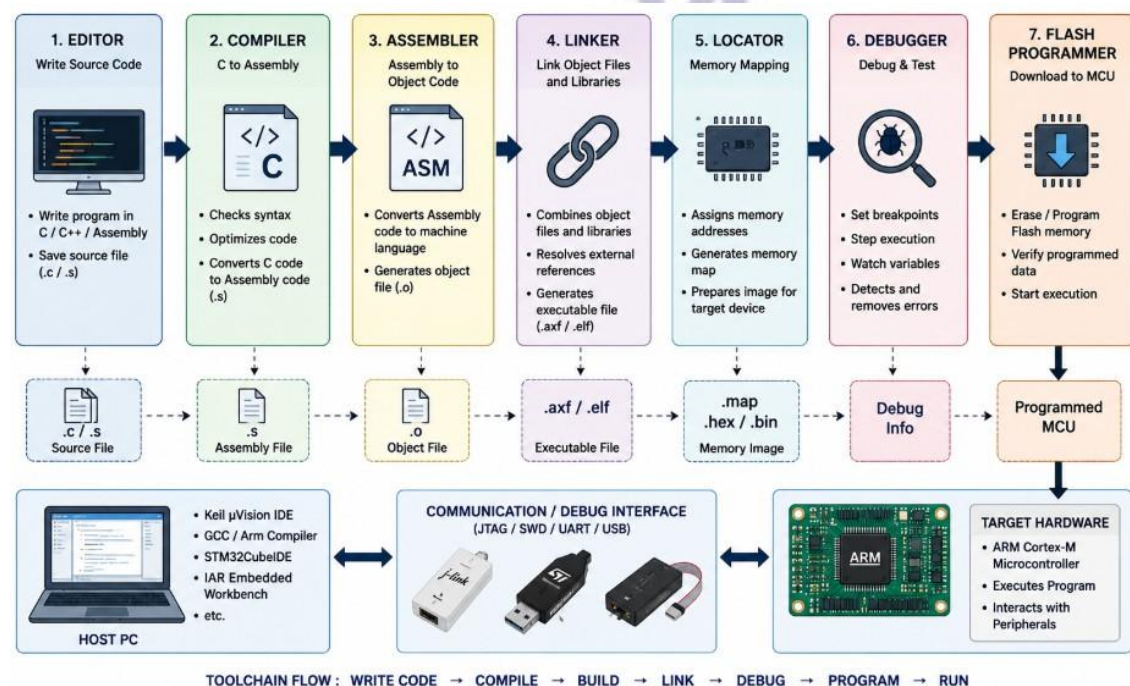
ARM (Advanced RISC Machine) processors are 32-bit Reduced Instruction Set Computing (RISC) processors widely used in embedded systems because of their high performance, low power consumption and compact instruction set. The LPC2148 microcontroller is based on the ARM7TDMI-S core and is extensively used in embedded applications such as industrial automation, robotics, IoT, consumer electronics and automotive systems.

1.5 ARM Development Toolchain

The ARM software development toolchain consists of the following components.

Tool	Function
Editor	Writing Embedded C/Assembly program
Compiler	Converts C program into Assembly language
Assembler	Converts Assembly program into Object code
Linker	Combines object files and libraries
Locator	Assigns memory locations to executable code
Debugger	Detects and removes software errors
Flash Programmer	Downloads program into microcontroller Flash memory

1.6 ARM Software Development Flow



1.7 Study of LPC2148 Evaluation Board

The LPC2148 evaluation board is based on the **NXP LPC2148 ARM7TDMI-S microcontroller**. It provides a complete hardware platform for learning embedded system programming and peripheral interfacing.

1.8 LPC2148 Specifications

Feature	Specification
Core	ARM7TDMI-S (32-bit RISC)
Maximum Clock Frequency	60 MHz
Flash Memory	512 KB
SRAM	40 KB
GPIO Pins	45
ADC	14-channel, 10-bit
DAC	1-channel, 10-bit
UART	2
SPI	2
I ² C	2
USB	USB 2.0 Full Speed
PWM	6 Channels
RTC	Available
Timers	Two 32-bit Timers

1.9 Study of Onboard Peripherals

1.10 Hardware Details

Board Level Information:

The following hardware features of the **Micro-A748 LPC2148 Evaluation Board** are to be studied before developing the Board Support Package (BSP).

Table 1.1: Features of Micro-A748 LPC2148 Evaluation Board

Sr. No.	Hardware Feature
1	Based on NXP LPC2148 ARM7TDMI-S microcontroller operating up to 60 MHz .
2	12 V DC Power Supply.
3	UART0 with USB-to-Serial interface and UART1 with RS-232 transceiver.
4	On-board 16 × 2 LCD module.
5	16-Key (4 × 4) Matrix Keypad.
6	Eight General Purpose LEDs.
7	Real-Time Clock (RTC) with battery backup.
8	EEPROM interfaced through I ² C bus.
9	SD/MMC Card interface using SPI protocol.
10	ZigBee Communication Module Interface.
11	10-bit On-chip ADC interfaced to external voltage source and temperature sensor.
12	10-bit Digital-to-Analog Converter (DAC).
13	Stepper Motor and DC Motor Driver using L293D IC.

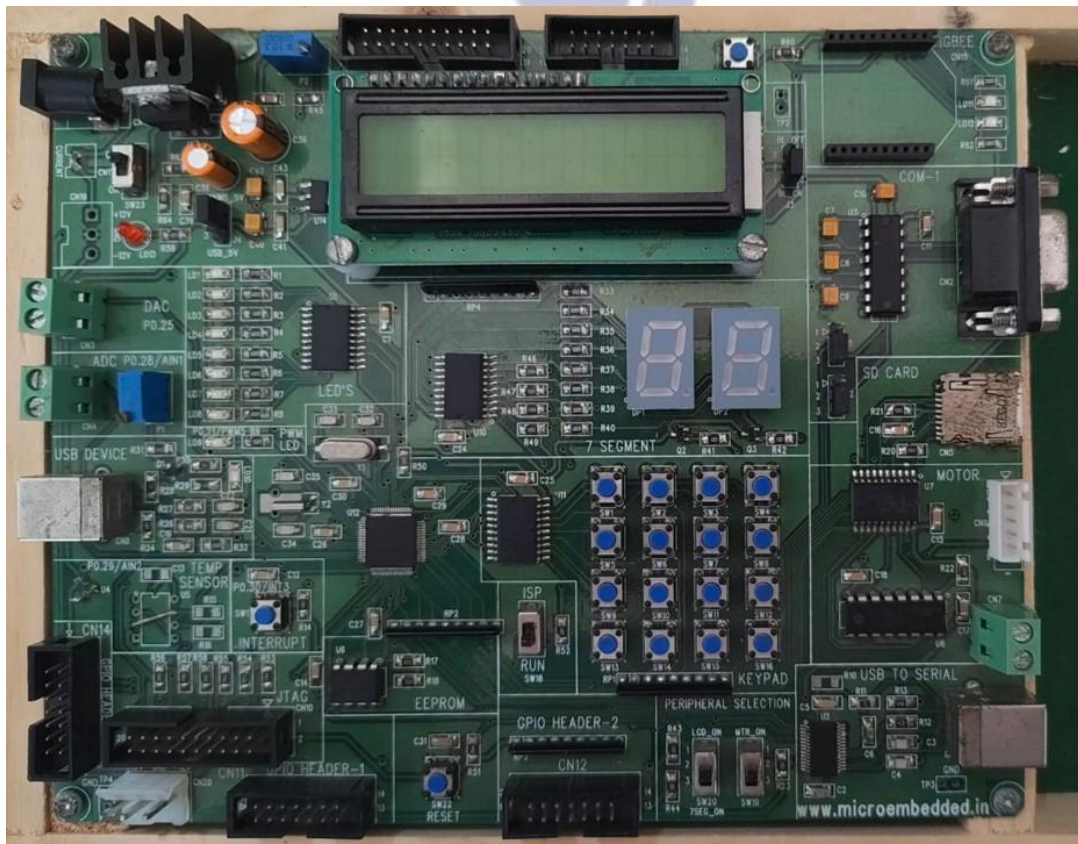


Table 1.2: Hardware Details: Mapping of the Hardware Pins with the Respective Interfacing Devices

LED Interfacing Details		
Device	LPC2148 Pin	Description
LED0	P1.24	Common Cathode LED 0
LED1	P1.25	Common Cathode LED 1
LED2	P1.26	Common Cathode LED 2
LED3	P1.27	Common Cathode LED 3
LED4	P1.28	Common Cathode LED 4
LED5	P1.29	Common Cathode LED 5
LED6	P1.30	Common Cathode LED 6
LED7	P1.31	Common Cathode LED 7
LCD Interfacing Details		
D0	P1.16	LCD Data Bit 0
D1	P1.17	LCD Data Bit 1
D2	P1.18	LCD Data Bit 2
D3	P1.19	LCD Data Bit 3
D4	P1.20	LCD Data Bit 4
D5	P1.21	LCD Data Bit 5
D6	P1.22	LCD Data Bit 6
D7	P1.23	LCD Data Bit 7
EN	P0.10	LCD Enable Signal
RS	P0.11	LCD Register Select
RW	P0.20	LCD Read/Write Control
Keypad Interfacing Details		
COL1	P0.12	Keypad Column 1
COL2	P0.13	Keypad Column 2
COL3	P0.14	Keypad Column 3
COL4	P0.15	Keypad Column 4
ROW1	P0.16	Keypad Row 1
ROW2	P0.17	Keypad Row 2
ROW3	P0.18	Keypad Row 3
ROW4	P0.19	Keypad Row 4
ADC and DAC Interfacing Details		
ADC	P0.28	ADC Input (ADC0.1)
DAC	P0.25	DAC Output

1.11 Procedure

1. Study the specifications of LPC2148.
2. Identify all onboard peripherals.
3. Install Keil μ Vision IDE.
4. Install LPC2148 Device Support Package.
5. Create a new ARM project.
6. Select LPC2148 microcontroller.
7. Configure startup files.
8. Build the project.
9. Observe compiler messages.
10. Generate HEX file.
11. Download program using Flash Magic.
12. Verify successful execution.

1.12 Observation Table

Parameter	Observation
ARM Board Used	LPC2148 Evaluation Board
IDE Used	Keil μ Vision 4
Controller Used	NXP LPC2148
Compiler Used	ARM C Compiler
Debugger Used	Keil Debugger
Build Status	Build Successful (0 Errors, 0 Warnings)
HEX File Generated	Yes
Download Status	Program Successfully Downloaded using Flash Magic

1.13 Result

The ARM development environment was successfully configured. The LPC2148 device was selected, the project was compiled successfully, the HEX file was generated, and the program was downloaded to the LPC2148 evaluation board using Flash Magic without any errors.

1.14 Precautions

- Select the correct LPC2148 device before creating the project.
- Verify compiler and debugger settings.
- Check USB connection before programming.
- Ensure correct COM port selection in Flash Magic.
- Build the project successfully before downloading.

1.15 Viva Questions

1. What is ARM7TDMI-S?
2. Why is LPC2148 widely used?
3. Explain ARM software toolchain.
4. What is CMSIS?
5. What is Flash Magic?
6. Explain ISP programming.
7. What are the onboard peripherals of LPC2148?
8. Why is ARM preferred over 8051?
9. Explain the function of JTAG.
10. Differentiate between Compiler and Linker.

1.16 Learning Outcomes

After completing this experiment, students will be able to

- Explain the ARM embedded software development process.
- Identify suitable hardware and software architecture for ARM-based embedded systems.
- Configure ARM development tools.
- Study the LPC2148 evaluation board and onboard peripherals.
- Develop the foundation for subsequent ARM programming and interfacing experiments.

1.17 REFERENCES

- LPC2148 User Manual **LPC2148**
- Andrew N. Sloss, Dominic Symes, Chris Wright, “ARM system developers guide, designing and optimizing system software”.

